

Question	Answer	Marks
11(a)(i)	packet/quantum/discrete amount of <u>energy</u>	M1
	of electromagnetic radiation	A1
11(a)(ii)	(maximum) energy of emitted electrons is independent of intensity or no emission of electrons below the threshold frequency regardless of intensity or no emission of electrons when photon energy is less than work function (energy) regardless of intensity	B1
11(b)	in darkness: conduction band empty so high resistance	B1
	in daylight: electrons in valence band absorb photons	B1
	in daylight: electrons 'jump' to conduction band	B1
	this leaves holes in valence band	B1
	more charge carriers in daylight so resistance decreases	B1

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Question	Answer	Marks
12(c)(i)	$\tau = \frac{1}{\omega} = \frac{1}{2\pi f}$	01